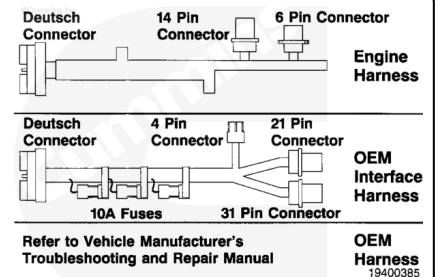


Trainings ▾

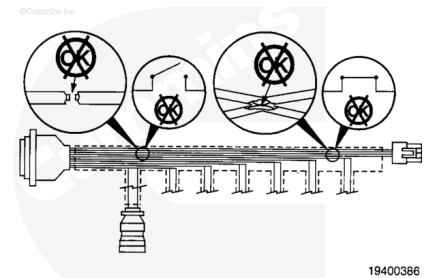
General Information

QSK-equipped engines use three separate wiring harnesses to control the engine and some of the vehicle operations.

1. Engine harness.
2. Original equipment manufacturer (OEM) interface harness.
3. OEM harness.



Replace a harness if there is an open or short circuit found under the protective covering of the harness body.

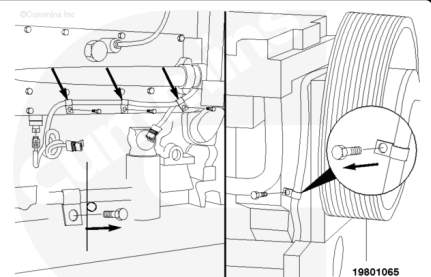


Remove

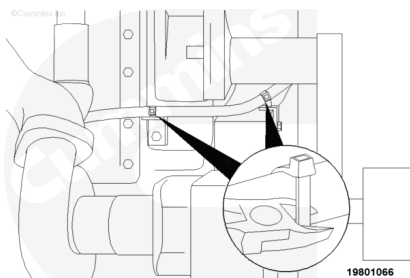
QSK19 Engines

Disconnect the engine harness from the coolant pressure sensor.

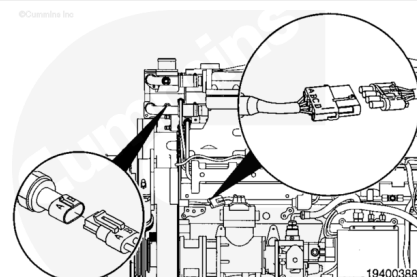
Remove the engine harness clamps.



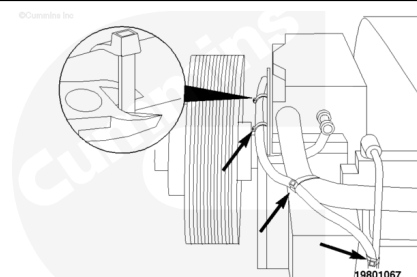
Cut the ties on the alternator bracket and front engine cover bracket.



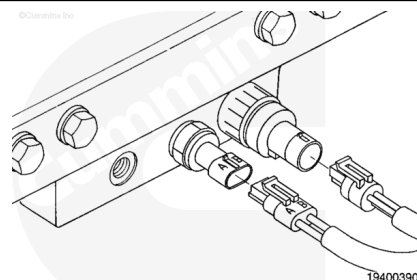
Disconnect the coolant temperature sensor and engine position sensor.



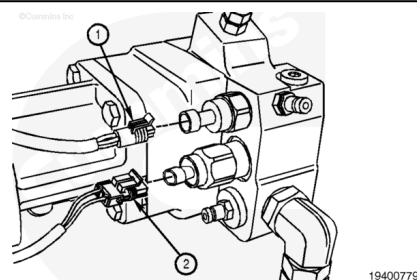
Cut the ties on the thermostat housing bracket, aftercooler tube, and fuel tube.



Disconnect the intake manifold temperature sensor and intake manifold pressure sensor.

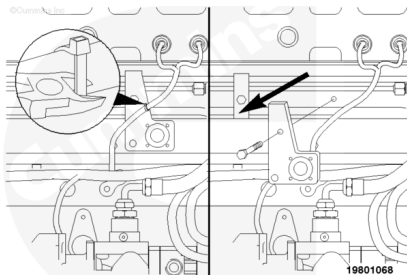


Disconnect fuel pump pressure sensor (1) and fuel pump actuator (2).

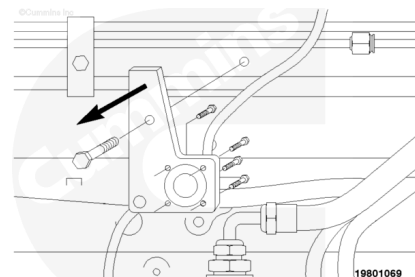


Cut the tie on the engine data link connector bracket.

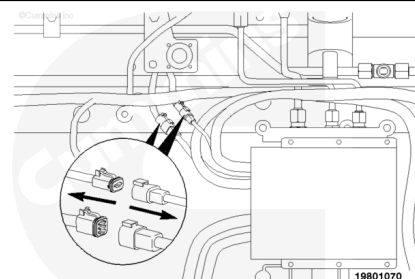
Remove the bracket.



Disconnect the data link connector from the harness support bracket by removing the four screws.

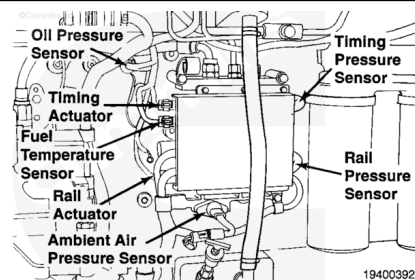


Disconnect the 2- and 6-pin Deutsch™ connectors.

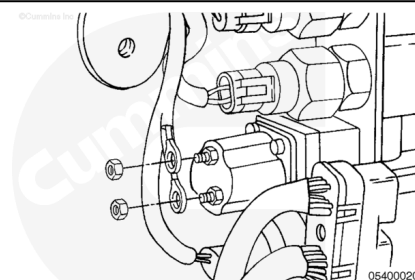


Disconnect the engine harness from the following:

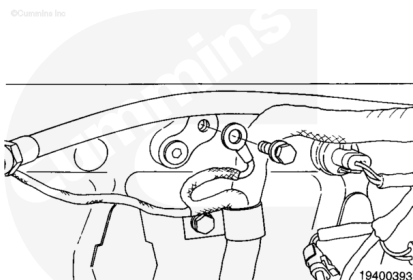
1. Oil pressure sensor.
2. Timing actuator.
3. Rail actuator.
4. Ambient air pressure sensor.
5. Rail pressure sensor.
6. Timing pressure sensor.
7. Fuel temperature sensor.



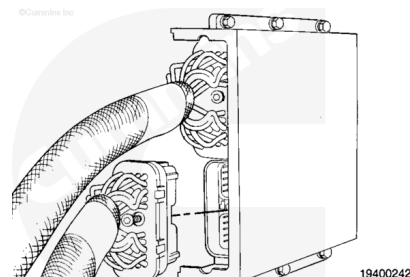
Disconnect the fuel shutoff control wires to the fuel shutoff solenoid.



Disconnect the engine block ground from the block.

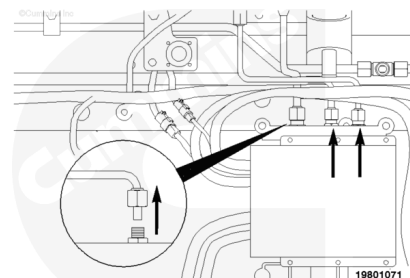


Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness connector from the engine control module (ECM).



Remove the fuel supply tubes from the top of the control valve body to gain access to the engine harness.

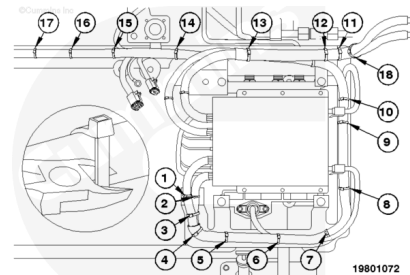
Refer to Procedure 006-024 in Section 6 (</qs3/pubsys2/xml/en/procedures/19/19-006-024.html>) in the Troubleshooting and Repair Manual, QSK19 Series Engines, Bulletin 3666098.



Cut the 17 ties holding the engine harness to the support bracket.

Cut the corner tie holding the engine harness to the support bracket.

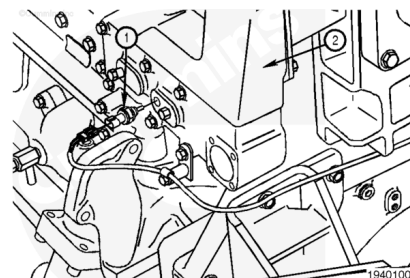
Remove the engine harness.



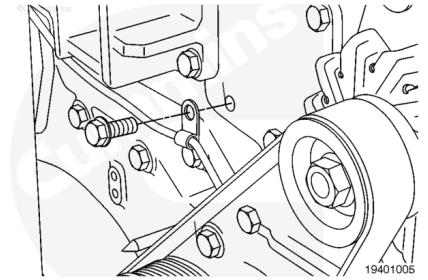
QSK23 Engines

Disconnect the engine harness from the coolant temperature sensor (1).

Remove the engine harness P-clip, spacer, and bracket mounted to thermostat housing (2).

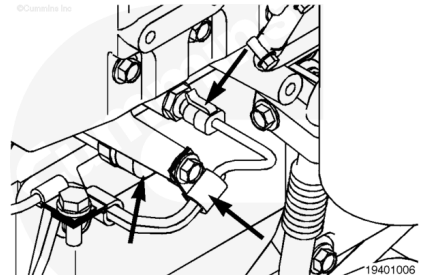


Remove the eight engine harness P-clips from the front of the engine, as shown.



Note : For power generation engines, these two sensors are located in the intake manifold on the right side of the fuel filters.

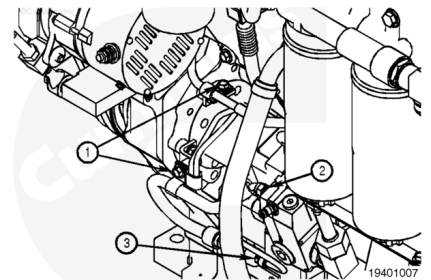
Disconnect the engine harness from the intake manifold temperature and pressure sensors, located on the right side of the fuel filters.



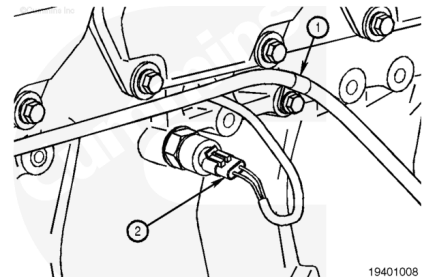
Remove the engine harness P-clip and spacer attached near the intake manifold pressure sensor.

Remove the engine harness P-clips (1) attached to the fuel pump mounting flange.

Disconnect the engine harness from the fuel pump actuator (2) and fuel pump pressure sensor (3) connectors.

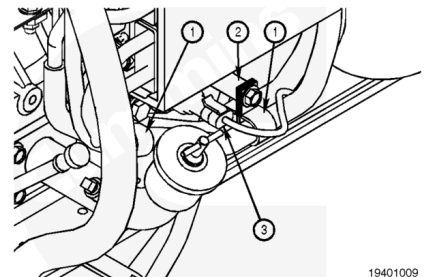


Remove the harness P-clip and the engine ground wire (1) and disconnect the oil pressure sensor (2).

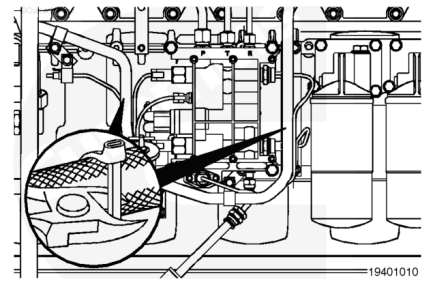


Remove the harness P-clips (1) and bracket (2) mounted to and just below the ECM near the ambient air pressure sensor (3).

Disconnect the ambient air pressure sensor (3).

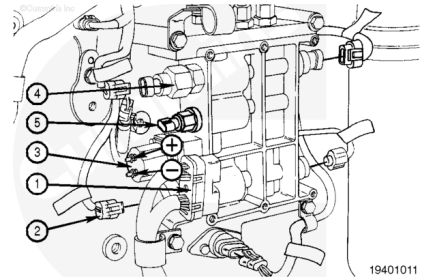


Cut the nine wire ties holding the engine harness to the support bracket.



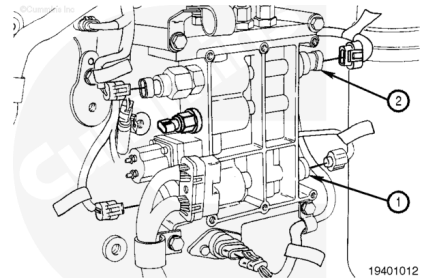
Disconnect the engine harness from the following:

1. ECM connector A
2. Fuel rail actuator
3. Fuel shutoff solenoid
4. Fuel timing actuator
5. Fuel temperature sensor.



Disconnect the engine harness from the following:

1. Rail pressure sensor
2. Timing pressure sensor.

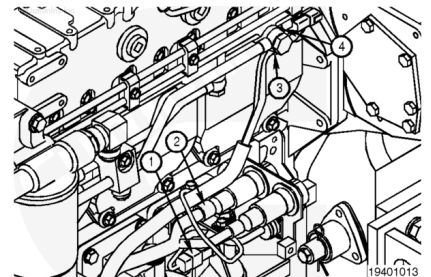


Note : Power generation wiring harness routing will be slightly different and the harness will have some different connectors.

Disconnect the data link (1) connector connecting the engine harness to the OEM harness.

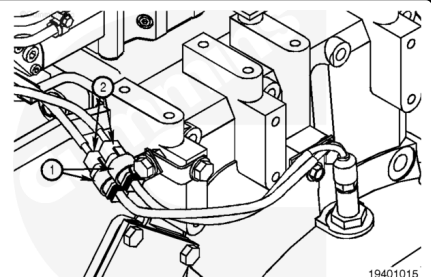
Remove the wiring P-clip (3) and intake heater connector bracket (4) located above the number 6 cam follower cover.

Remove the P-clip (2) near the lower right corner of the number 5 cam follower cover.



Remove the P-clips (1) to the rear of the intake manifold and the rear of the cylinder block.

Disconnect the engine speed sensor connectors (2) and remove the engine harness.

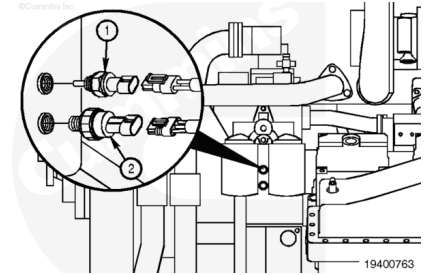


QSK45 Engines

Note : The art shown in the following steps is based on the QSK60 engine. The QSK45 engine configuration is similar.

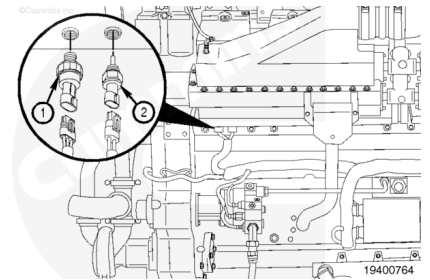
Disconnect the engine harness from the coolant temperature sensor (1).

Disconnect the coolant pressure sensor (2) (CENSE™ equipped engines **only**).

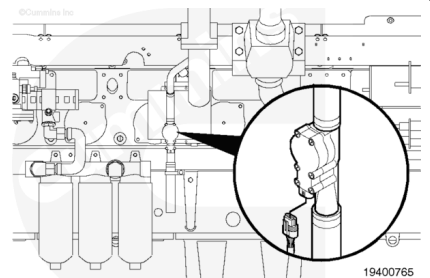


Disconnect the engine harness from the intake manifold air pressure sensor (1).

Disconnect the engine harness from the intake manifold air temperature sensor (2).

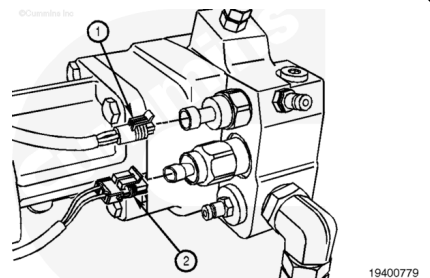


Disconnect the engine harness from the blowby pressure sensor (CENSE™ equipped engines **only**).



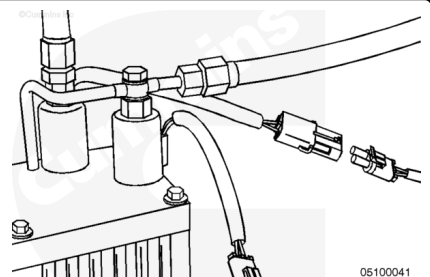
Disconnect the engine harness from the fuel pump pressure sensor (1).

Disconnect the engine harness from the fuel pump actuator (2).



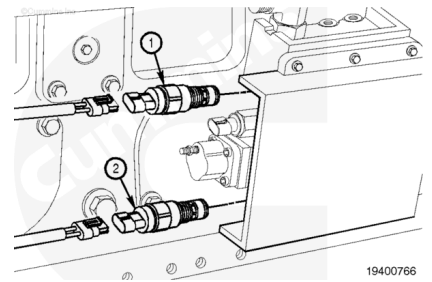
If equipped, disconnect the Centinel™ burn valve solenoid from the engine harness.

If equipped, disconnect the Centinel™ make-up valve solenoid from the engine harness.

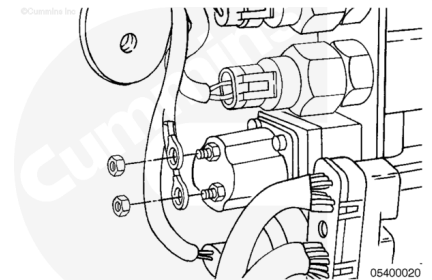


Disconnect the engine harness from the two timing actuators (1).

Disconnect the engine harness from the rail actuator (2).

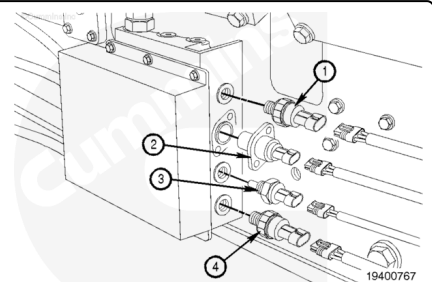


Disconnect the fuel shutoff control wires and ground wires to the fuel shutoff solenoid.



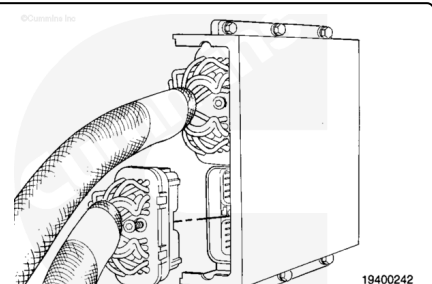
Disconnect the engine harness from the following:

1. Timing pressure sensor.
2. Ambient air pressure sensor.
3. Fuel temperature sensor.
4. Rail pressure sensor.

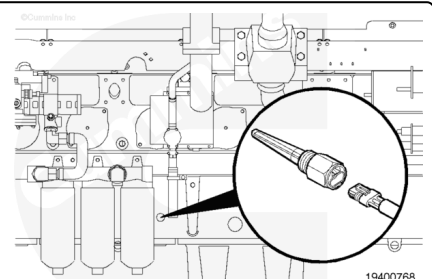


Disconnect the engine harness from the ECM.

Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness connector from the ECM.

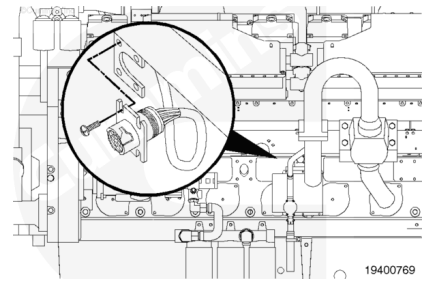


If equipped, disconnect the engine harness from the oil level sensor.



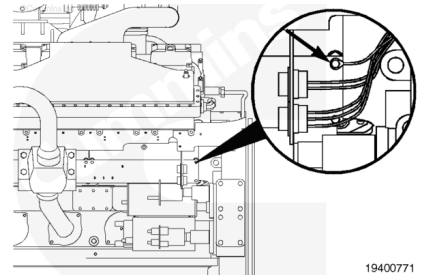
Disconnect the 9-pin data link connector from the engine harness support bracket.

- Remove the four capscrews.
- Slide the 9-pin data link connector out of the mounting slot in the engine harness support bracket.



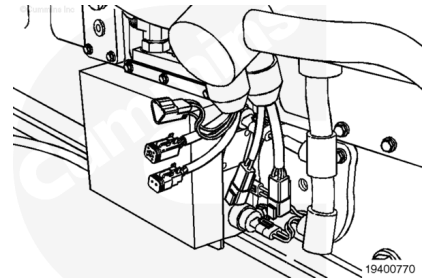
19400769

Disconnect the engine block ground.



19400771

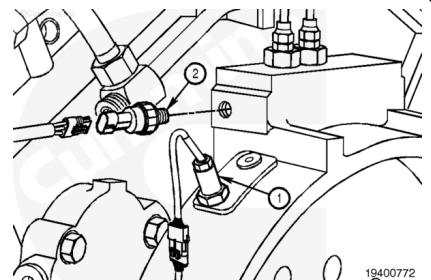
Disconnect the three data link connectors.



19400770

Disconnect the engine harness from the engine speed sensor (1).

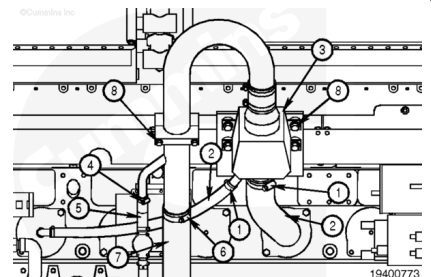
Disconnect the engine harness from the oil pressure sensor (2).



19400772

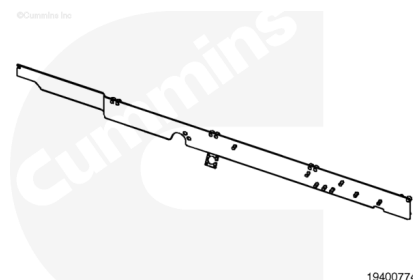
Remove the breather box assembly.

- Loosen the three hose clamps (1) and disconnect the three hoses (2) on the bottom of the breather box (3).
- Loosen the hose clamp (4) and disconnect the blowby pressure sensor hose (5).
- Loosen the hose clamp (6) and disconnect the lower breather hose (7).
- Remove the six attaching bolts (8) and remove the breather box assembly.

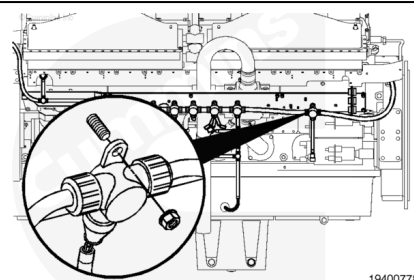


19400773

Remove the wire clamps from the harness support bracket.
Remove the engine harness from the clamps and bracket.



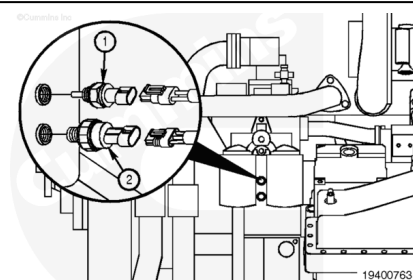
Remove the wire clamps from the harness support bracket.
Remove the engine harness from the clamps and bracket.



QSK60 Engines

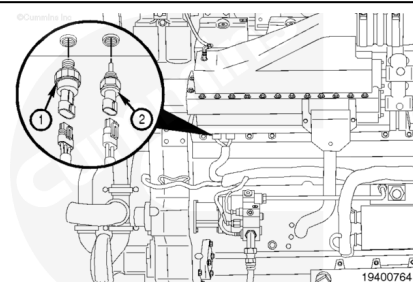
Disconnect the engine harness from the coolant temperature sensor (1).

Disconnect the coolant pressure sensor (2) (CENSE™ equipped engines **only**).

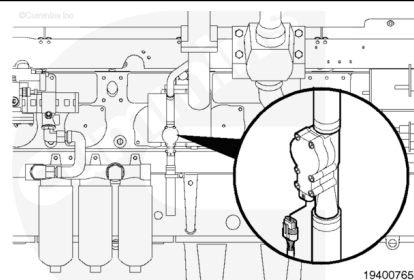


Disconnect the engine harness from the intake manifold air pressure sensor (1).

Disconnect the engine harness from the intake manifold air temperature sensor (2).

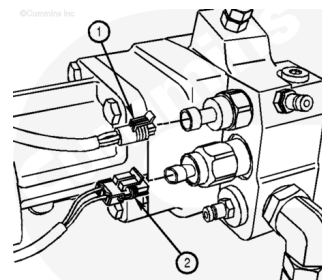


Disconnect the engine harness from the blowby pressure sensor (CENSE™ equipped engines **only**).



Disconnect the engine harness from the fuel pump pressure sensor (1).

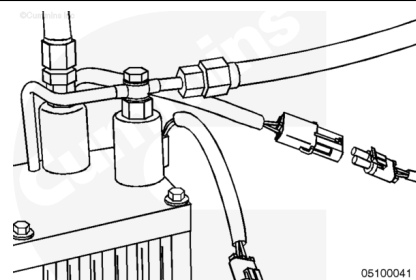
Disconnect the engine harness from the fuel pump actuator (2).



19400779

If equipped, disconnect the Centinel™ burn valve solenoid from the engine harness.

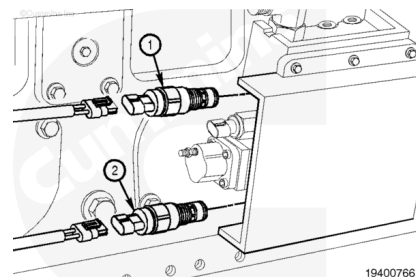
If equipped, disconnect the Centinel™ make-up valve solenoid from the engine harness.



05100041

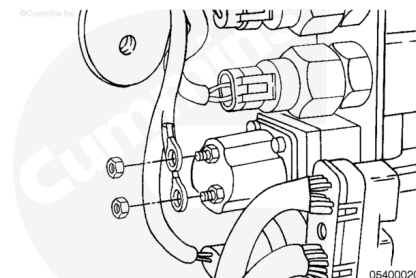
Disconnect the engine harness from the two timing actuators (1).

Disconnect the engine harness from the rail actuator (2).



19400766

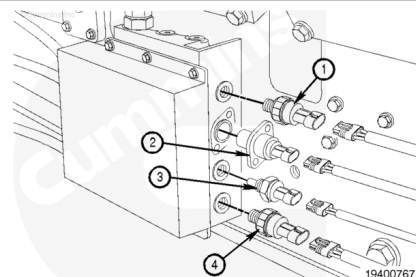
Disconnect the fuel shutoff control wires and ground wires from the fuel shutoff solenoid.



05400020

Disconnect the engine harness from the following:

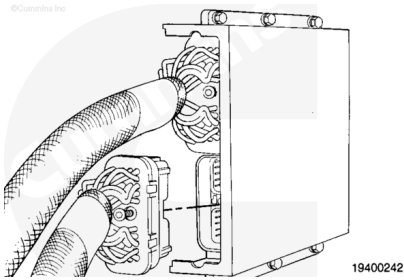
1. Timing pressure sensor.
2. Ambient air pressure sensor.
3. Fuel temperature sensor.
4. Rail pressure sensor.



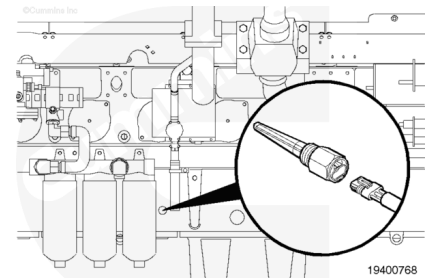
19400767

Disconnect the engine harness from the ECM.

Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness connector from the ECM.

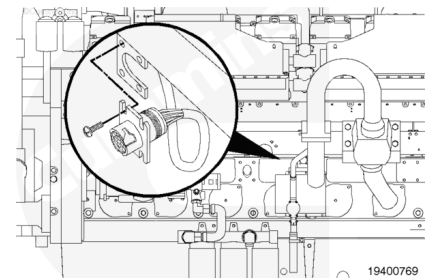


If equipped, disconnect the engine harness from the oil level sensor.

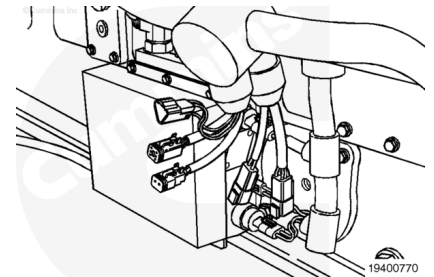


Disconnect the 9-pin data link connector from the engine harness support bracket.

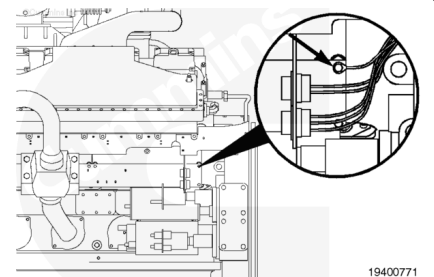
- Remove the four capscrews.
- Slide the 9-pin data link connector out of the mounting slot in the engine harness support bracket.



Disconnect the three data link connectors.

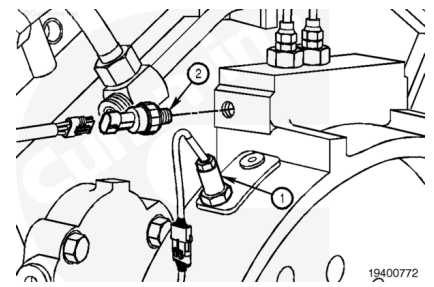


Disconnect the engine block ground.



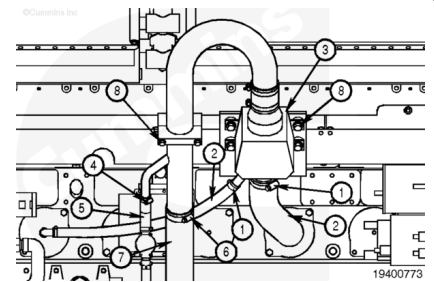
Disconnect the engine harness from the engine speed sensor (1).

Disconnect the engine harness from the oil pressure sensor (2).



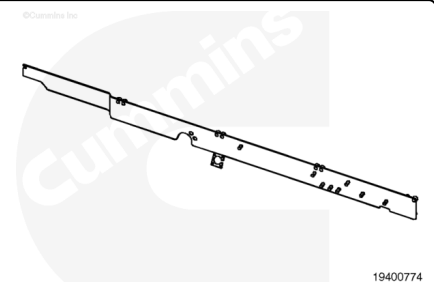
Remove the breather box assembly.

- Loosen the three hose clamps (1) and disconnect the three hoses (2) on the bottom of the breather box (3).
- Loosen the hose clamp (4) and disconnect the blowby pressure sensor hose (5).
- Loosen the hose clamp (6) and disconnect the lower breather hose (7).
- Remove the six attaching bolts (8) and remove the breather box assembly.



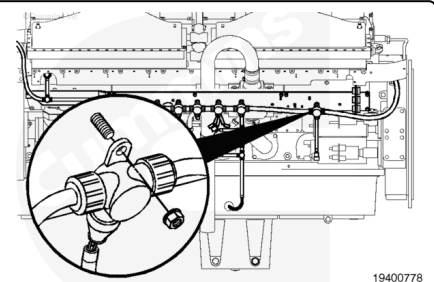
Remove the harness support bracket from the engine.

- Double-check that all engine harness connectors are disconnected and untangled.
- Carefully remove the nine attaching bolts from the support bracket.
- Slowly pull the support bracket from the engine, making sure the harness is pulled from the engine without binding or tangling.



Remove the wire clamps from the harness support bracket.

Remove the engine harness from the clamps and bracket.

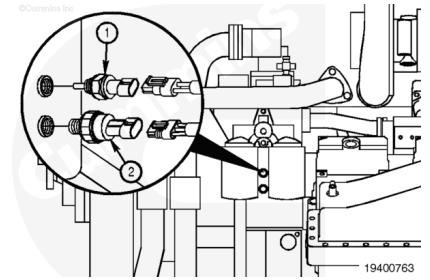


QSK78 Engines

Note : The art shown in the following steps is based on the QSK60 engine. The QSK78 engine configuration is similar.

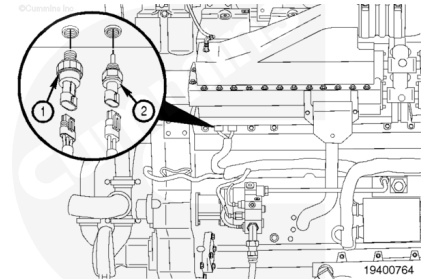
Disconnect the engine harness from the coolant temperature sensor (1).

Disconnect the coolant pressure sensor (2) (CENSE™-equipped engines **only**).

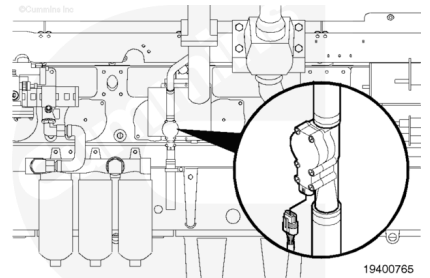


Disconnect the engine harness from the intake manifold air pressure sensor (1).

Disconnect the engine harness from the intake manifold air temperature sensor (2).

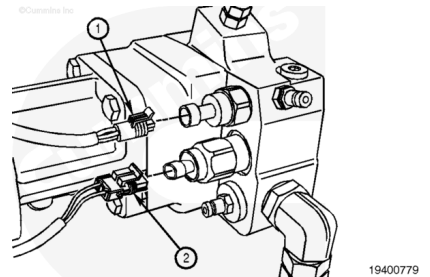


Disconnect the engine harness from the blowby pressure sensor (CENSE™ equipped engines **only**).



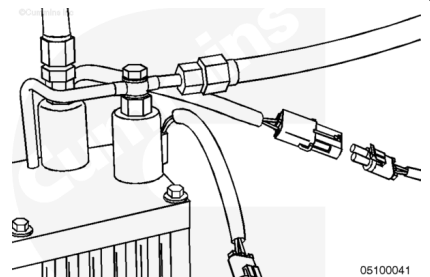
Disconnect the engine harness from the fuel pump pressure sensor (1).

Disconnect the engine harness from the fuel pump actuator (2).



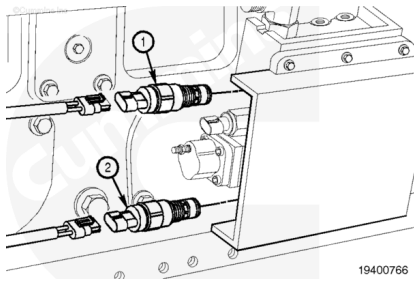
If equipped, disconnect the Centinel™ burn valve solenoid from the engine harness.

If equipped, disconnect the Centinel™ make-up valve solenoid from the engine harness.

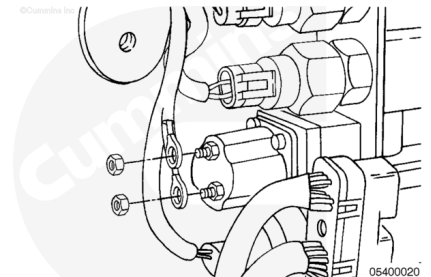


Disconnect the engine harness from the two timing actuators (1).

Disconnect the engine harness from the rail actuator (2).

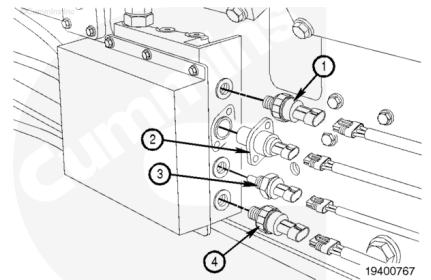


Disconnect the fuel shutoff control wires and ground to the fuel shutoff solenoid.



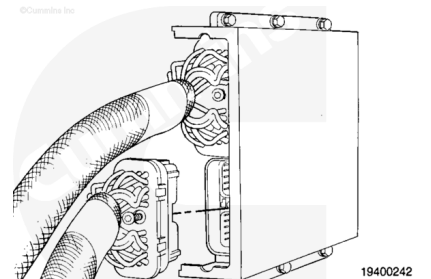
Disconnect the engine harness from the following:

1. Timing pressure sensor.
2. Ambient air pressure sensor.
3. Fuel temperature sensor.
4. Rail pressure sensor.

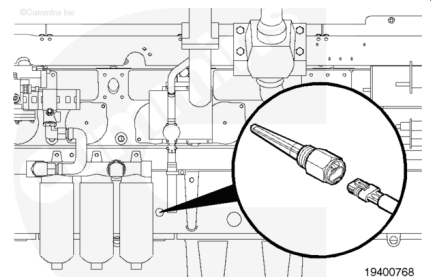


Disconnect the engine harness from the ECM.

Use a 4 mm [5/32 in] hex head wrench to disconnect the engine harness connector from the ECM.



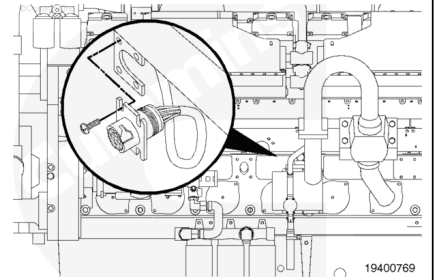
If equipped, disconnect the engine harness from the oil level sensor.



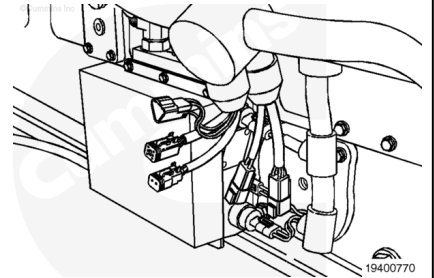
Disconnect the 9-pin data link connector from the engine harness support bracket:

1. Remove four capscrews.

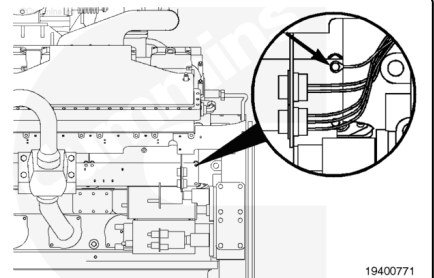
2. Slide the 9-pin data link connector out of the mounting slot in the engine harness support bracket.



Disconnect the three data link connectors.

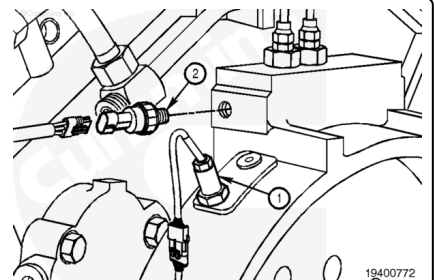


Disconnect the engine block ground.



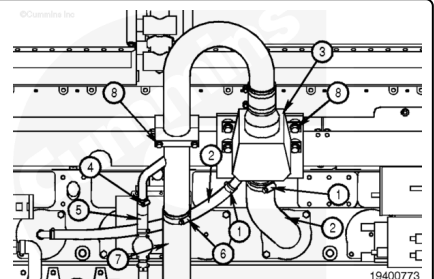
Disconnect the engine harness from the engine speed sensor (1).

Disconnect the engine harness from the oil pressure sensor (2).



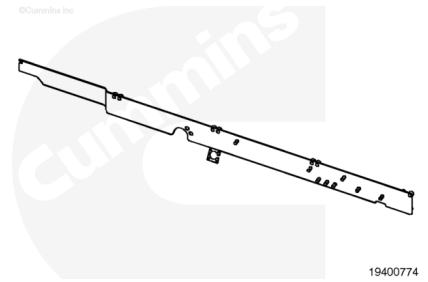
Remove the breather box assembly.

- Loosen the three hose clamps (1) and disconnect the three hoses (2) on the bottom of the breather box (3).
- Loosen the hose clamp (4) and disconnect the blowby pressure sensor hose (5).
- Loosen the hose clamp (6) and disconnect the lower breather hose (7).
- Remove the six attaching bolts (8) and remove the breather box assembly.

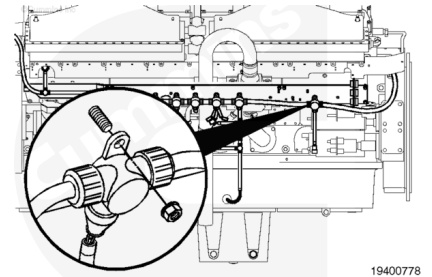


Remove the harness support bracket from the engine.

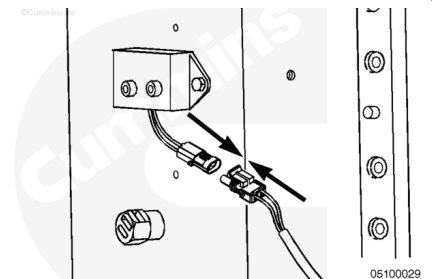
- Double-check that all engine harness connectors are disconnected and untangled.
- Carefully remove the nine attaching bolts from the support bracket.
- Slowly pull the support bracket from the engine, making sure the harness is pulled from the engine without binding or tangling.



Remove the wire clamps from the harness support bracket.
Remove the engine harness from the clamps and bracket.



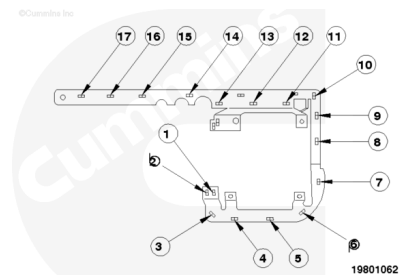
Disconnect the engine harness from the Centinel™ makeup tank oil level sensor inline connector.



Install

QSK19 Engines

Remove the engine harness from the clamps and bracket.

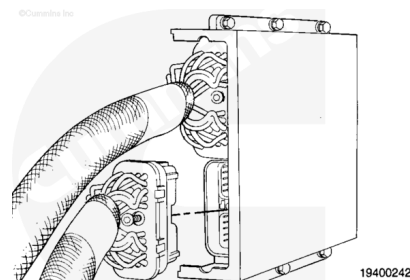


Connect the engine harness connector to the ECM.

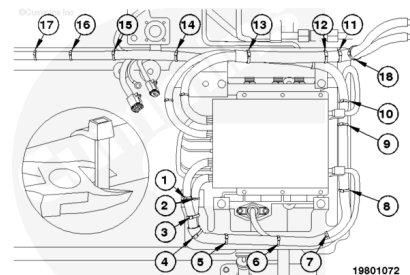
Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.

Torque Value: 2.8 n·m [25 in-lb]

Note : Do not over-torque as connector damage can occur.

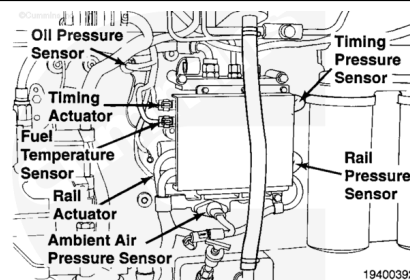


Fasten the harness to the harness support brackets using wire ties.



Connect the engine harness to the following:

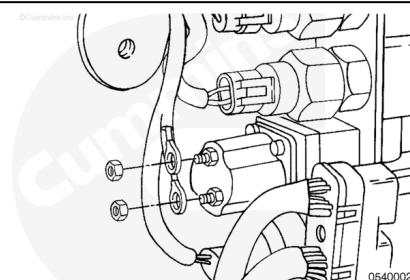
1. Oil pressure sensor.
2. Timing actuator.
3. Rail actuator
4. Ambient air (barometric) pressure sensor.
5. Rail pressure sensor.
6. Timing pressure sensor.
7. Fuel temperature sensor.



Connect the fuel shutoff control wires and ground wires to the fuel shutoff solenoid.

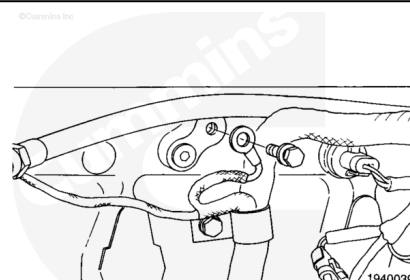
Tighten the retaining nuts.

Torque Value: 3 n•m [27 in-lb]

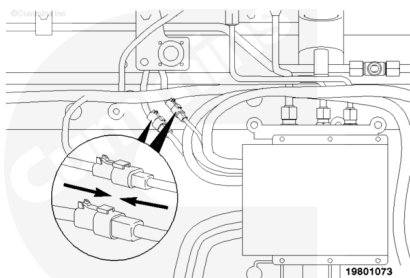


Connect the engine block ground to the engine block.

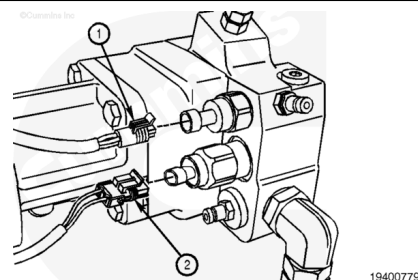
Torque Value: 20 n•m [177 in-lb]



Connect the 2- pin and 6-pin Deutsch™ connectors.

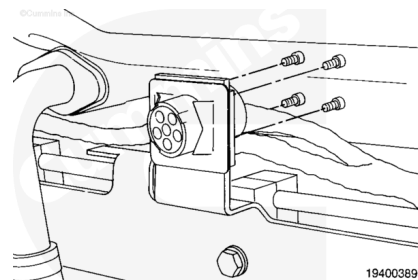


Connect the fuel pump pressure sensor (1) and actuator (2).



Connect the data link connector to the harness support bracket.

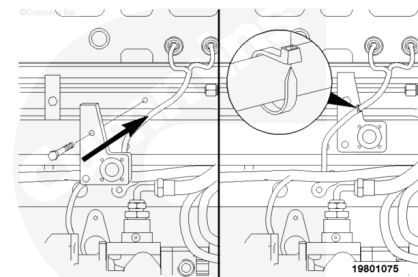
Torque Value: 3 n•m [27 in-lb]



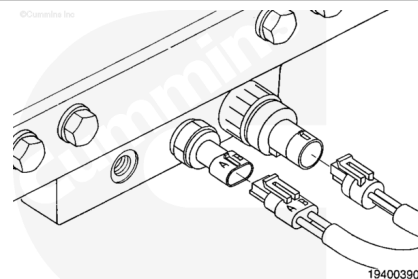
Install the data link connector support bracket.

Torque Value: 4 n•m [35 in-lb]

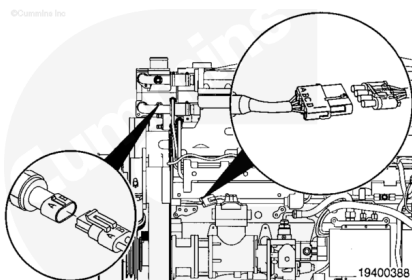
Fasten the harness for the intake manifold temperature sensor and intake manifold pressure sensor to the data link support bracket.



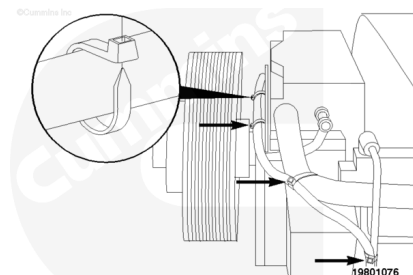
Connect the wiring harness to the intake manifold temperature sensor and intake manifold pressure sensor.



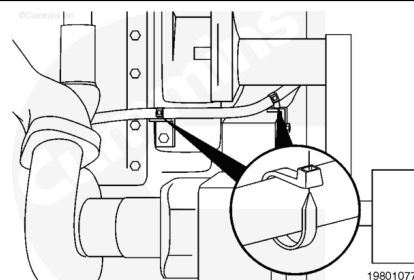
Connect the coolant temperature sensor and engine position sensor.



Fasten the harness to the fuel tube, aftercooler tube, and thermostat housing.



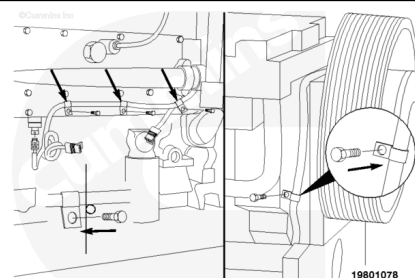
Fasten the harness to the front engine cover bracket and alternator bracket.



Connect the engine harness to the coolant pressure sensor.

Install the engine harness clamps.

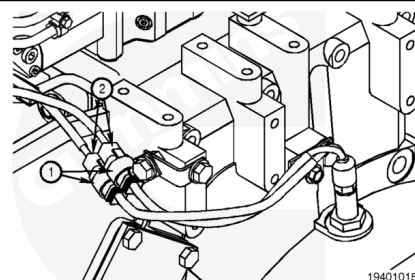
Torque Value: 20 N·m [177 in-lb]



QSK23 Engines

Install the engine harness and connect the engine speed sensor connectors (2).

Install the P-clips (1) to the rear of the intake manifold and the rear of the cylinder block.

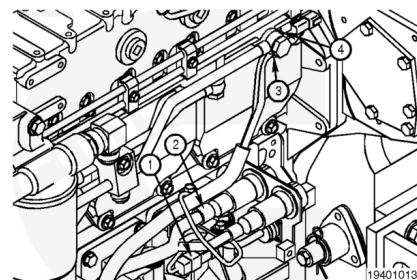


Note : Power generation wiring harness routing will be slightly different and the harness will have some different connectors.

Install the P-clip (2) near the lower right corner of the number 5 cam follower cover.

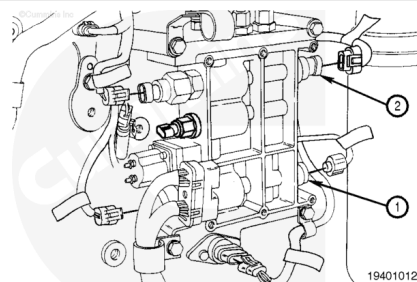
Install the wiring P-clip (3) and intake heater connector bracket (4) located above the number 6 cam follower cover.

Connect the data link (1) connector connecting the engine harness to the OEM harness.



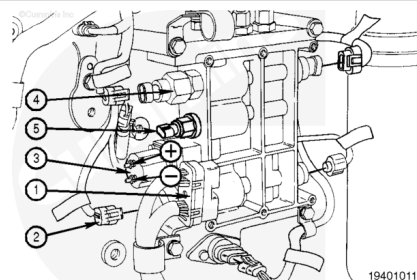
Connect the following to the engine harness:

1. Rail pressure sensor.
2. Timing pressure sensor.



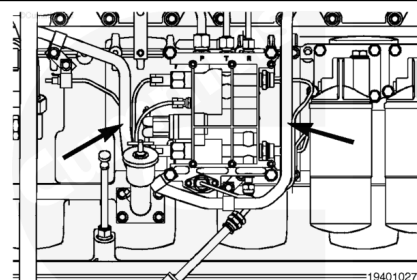
Connect the following to the engine harness:

1. ECM connector A.
2. Fuel rail actuator.
3. Fuel shutoff solenoid.
4. Fuel timing actuator.
5. Fuel temperature sensor.



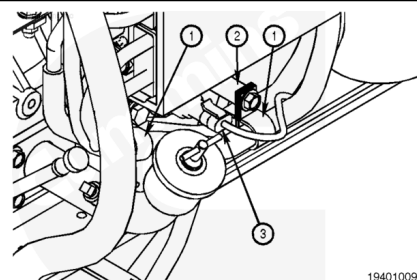
Make sure the positive (+) and negative (-) terminals on the fuel shutoff solenoid are correct.

Install the nine wire ties holding the engine harness to the support bracket.

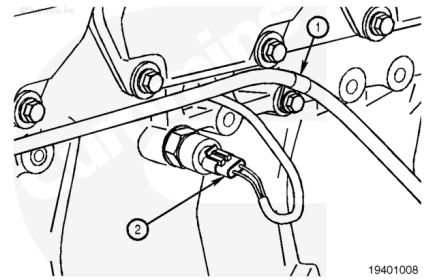


Connect the engine harness to the ambient pressure sensor (3).

Install the harness P-clips (1) and bracket (2) mounted to and just below the ECM near the ambient pressure sensor (3).

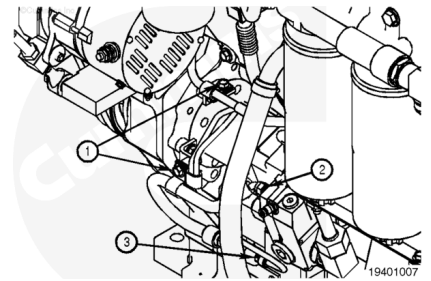


Install the harness P-clip and engine ground wire (1) and connect the oil pressure sensor (2).



Connect the harness to the fuel pump actuator (2) and fuel pump pressure sensor (3).

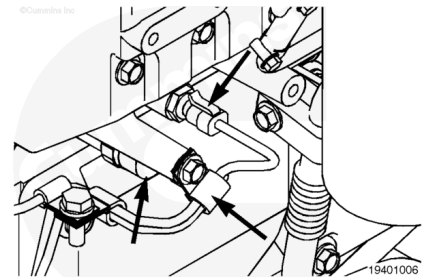
Install the engine harness P-clips (1) attached to fuel pump mounting flange.



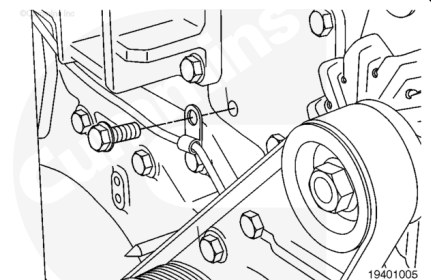
Note : For power generation engines, these two sensors are located in the intake manifold on the right side of the fuel filters.

Install the engine harness P-clip and spacer attached near the intake manifold pressure sensor.

Connect the engine harness to the intake manifold temperature and pressure sensors, located on the intake manifold on the right hand side of the fuel filters.

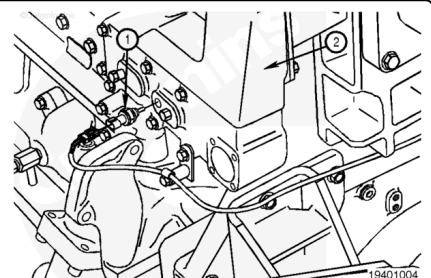


Install the eight engine harness P-clips on the front of the engine, as shown.



Install the engine harness P-clip, spacer, and bracket mounted to the thermostat housing (2).

Connect the engine harness to the coolant temperature sensor (1).

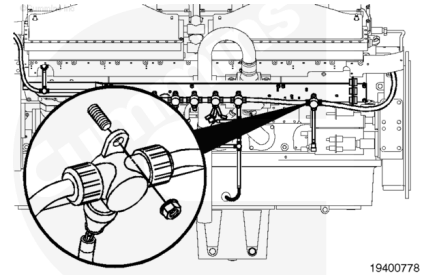


QSK45 Engines

Note : The art shown in the following steps is based on the QSK60 engine. The QSK45 engine configuration is similar.

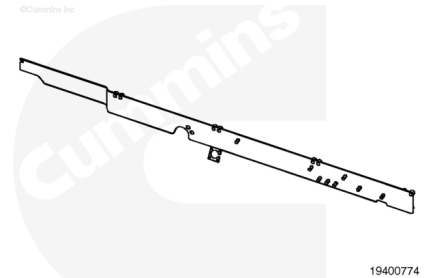
Place the engine harness on the harness support brackets.

Fasten the engine harness to the harness support brackets with the wire clamps.



Install the engine harness bracket with nine bolts.

Torque Value: 45 n·m [33 ft-lb]

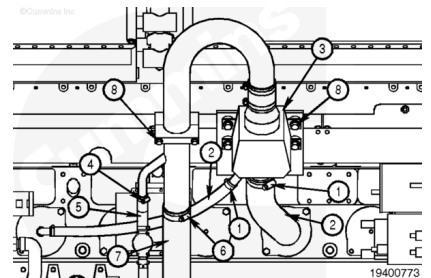


Install the breather box assembly.

- Connect the breather box to the harness support bracket with six bolts (8).

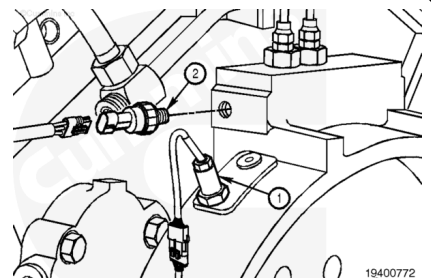
Torque Value: 45 n·m [33 ft-lb]

- Connect the lower breather hose (7) with the hose clamp (6).
- Connect the blowby pressure sensor hose (5) with the hose clamp (4).
- Connect the three hoses (2) on the bottom of the breather box (3) with the three hose clamps (1).



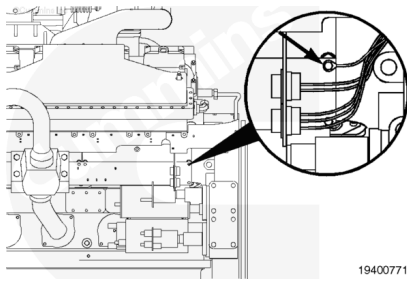
Connect the engine speed sensor (1) to the engine harness connector.

Connect the oil pressure sensor (2) to the engine harness connector.



Connect the engine block ground.

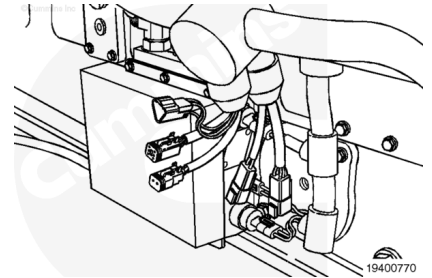
Torque Value: 20 n·m [177 in-lb]



19400771

Connect the three data link connectors.

Connect the engine harness to the fuel supply pump pressure and temperature sensors.

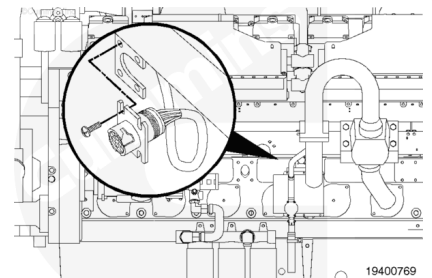


19400770

Connect the 9-pin data link connector to the engine harness support bracket.

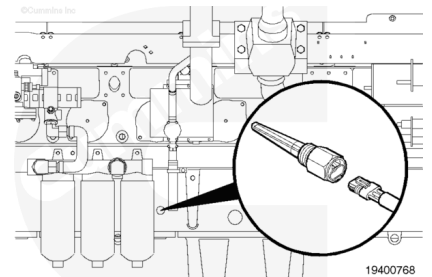
- Slide the 9-pin connector into the engine harness support bracket connector slot.
- Secure the 9-pin connector with four screws.

Torque Value: 1 n•m [9 in-lb]



19400769

Connect the oil level sensor to the engine harness.



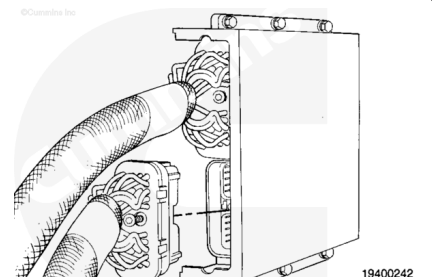
19400768

Connect the engine harness ECM connector to the ECM.

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.

Torque Value: 2.8 n•m [25 in-lb]

Note : Do not over torque as connector damage can occur.

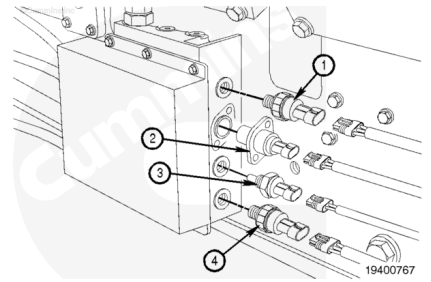


19400242

Connect the engine harness to the following:

- Timing pressure sensor (1).

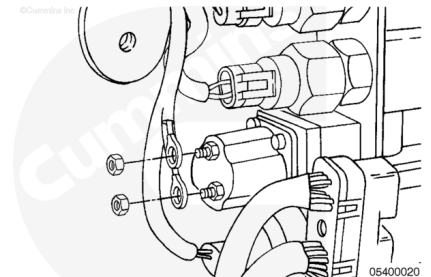
- Ambient air pressure sensor (2).
- Fuel temperature sensor (3).
- Rail pressure sensor (4).



Connect the fuel shutoff control wires and ground wires to the fuel shutoff solenoid.

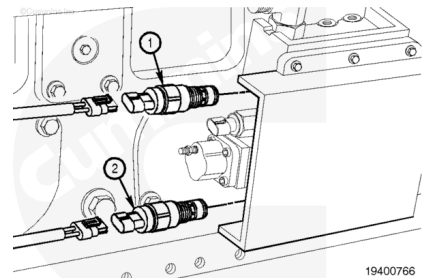
Tighten the retaining nuts.

Torque Value: 3 n•m [27 in-lb]



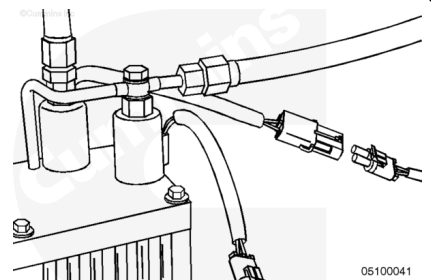
Connect the timing actuators (1) to the engine harness.

Connect the rail actuator (2) to the engine harness.



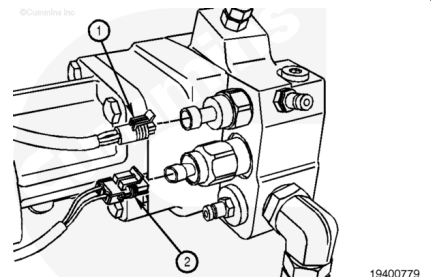
If equipped, connect the Centinel™ burn valve solenoid to the engine harness.

If equipped, connect the Centinel™ make-up valve solenoid to the engine harness.

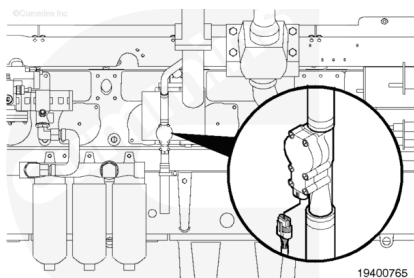


Connect the fuel pump pressure sensor (1) to the engine harness.

Connect the fuel pump actuator (2) to the engine harness.

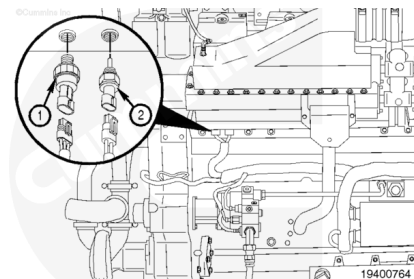


If CENSE™ equipped, connect the blowby pressure sensor to the engine harness.



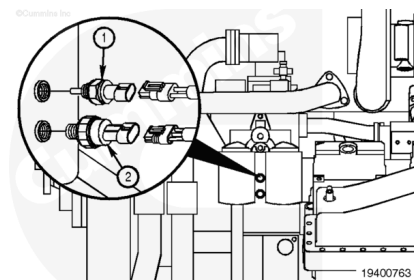
Connect the intake manifold air pressure sensor (1) to the engine harness.

Connect the intake manifold air temperature sensor (2) to the engine harness.



Connect the coolant temperature sensor (1) to the engine harness.

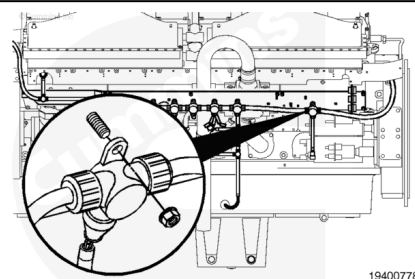
If CENSE™ equipped, connect the coolant pressure sensor (1) to the engine harness.



QSK60 Engines

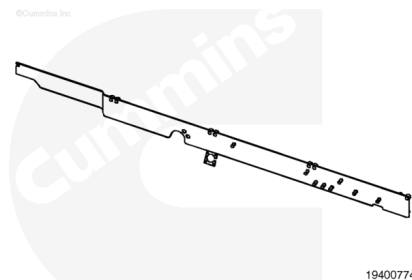
Place the engine harness on the harness support brackets.

Fasten the engine harness to the harness support brackets with the wire clamps.



Install the engine harness bracket with nine bolts.

Torque Value: 45 n•m [33 ft-lb]

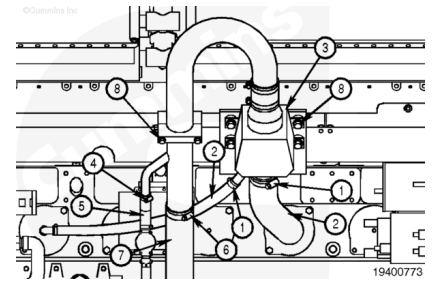


Install the breather box assembly.

- Connect the breather box to the harness support bracket with six bolts (8).

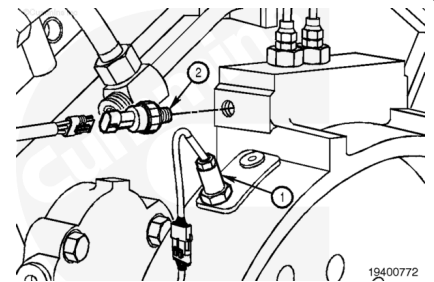
Torque Value: 45 n•m [33 ft-lb]

- Connect the lower breather hose (7) with the hose clamp (6).
- Connect the blowby pressure sensor hose (5) with the hose clamp (4).
- Connect the three hoses (2) on the bottom of the breather box (3) with the three hose clamps (1).



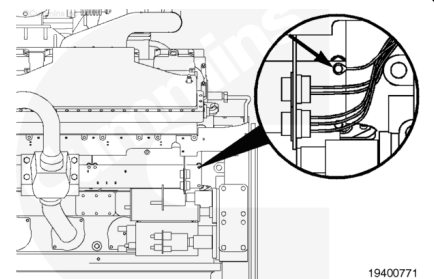
Connect the engine speed sensor (1) to the engine harness connector.

Connect the oil pressure sensor (2) to the engine harness connector.

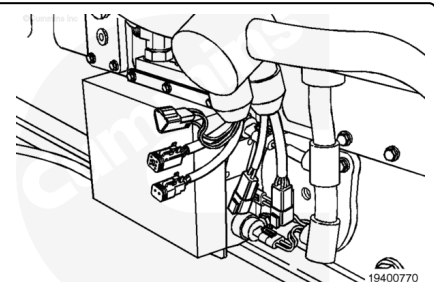


Connect the engine block ground.

Torque Value: 20 n•m [177 in-lb]



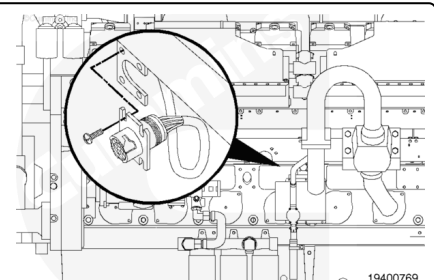
Connect the three data link connectors.



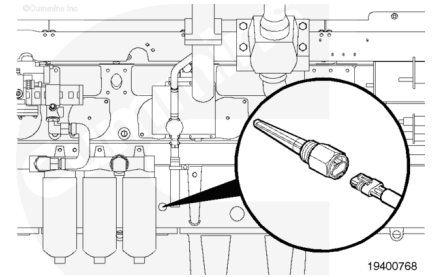
Connect the 9-pin data link connector to the engine harness support bracket.

- Slide the 9-pin connector into the engine harness support bracket connector slot.
- Secure the 9-pin connector with four screws.

Torque Value: 1 n•m [9 in-lb]



Connect the oil level sensor to the engine harness.

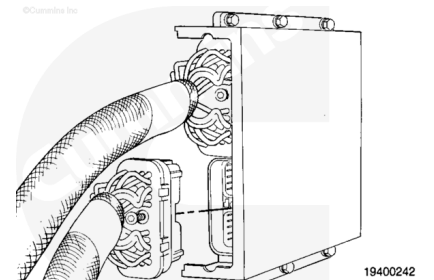


Connect the engine harness ECM connector to the ECM.

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew

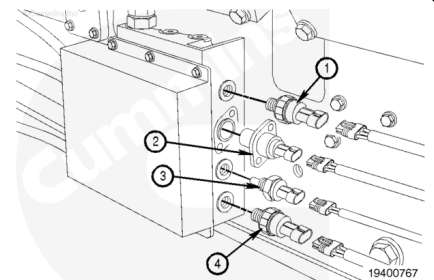
Torque Value: 2.8 n•m [25 in-lb]

Note : Do not over-torque as connector damage can occur.



Connect the engine harness to the following:

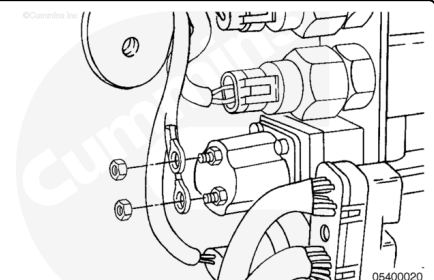
- Timing pressure sensor (1).
- Ambient air pressure sensor (2).
- Fuel temperature sensor (3).
- Rail pressure sensor (4).



Connect the fuel shutoff control wires and ground wires to the fuel shutoff solenoid.

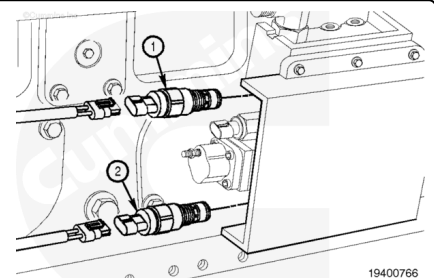
Tighten the retaining nuts.

Torque Value: 3 n•m [27 in-lb]



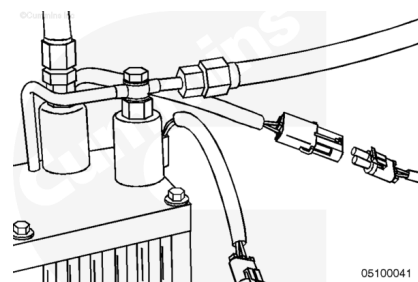
Connect the timing actuators (1) to the engine harness.

Connect the rail actuator (2) to the engine harness.



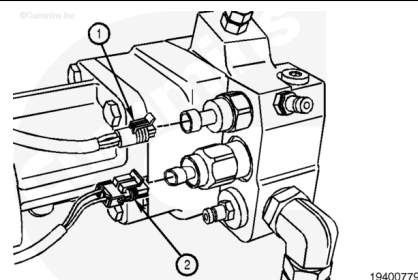
If equipped, connect the Centinel™ burn valve solenoid to the engine harness.

If equipped, connect the Centinel™ make-up valve solenoid to the engine harness.

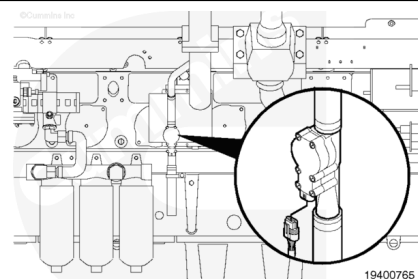


Connect the fuel pump pressure sensor (1) to the engine harness.

Connect the fuel pump actuator (2) to the engine harness.

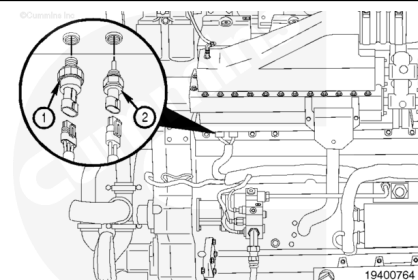


If CENSE™ equipped, connect the blowby pressure sensor to the engine harness.



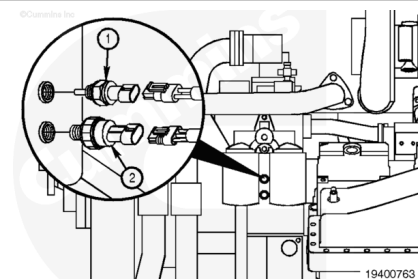
Connect the intake manifold air pressure sensor (1) to the engine harness.

Connect the intake manifold air temperature sensor (2) to the engine harness.



Connect the coolant temperature sensor (1) to the engine harness.

If CENSE™ equipped, connect the coolant pressure sensor (1) to the engine harness.

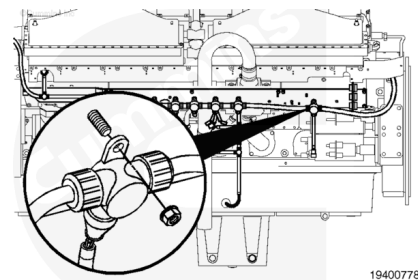


QSK78 Engines

Note : The art shown in the following steps is based on the QSK60 engine. The QSK78 engine configuration is similar.

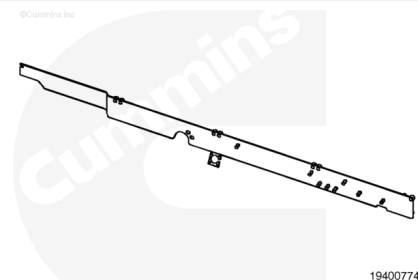
Place the engine harness on the harness support brackets.

Fasten the engine harness to the harness support brackets with the wire clamps.



Install the engine harness bracket with nine bolts.

Torque Value: 45 n·m [33 ft-lb]

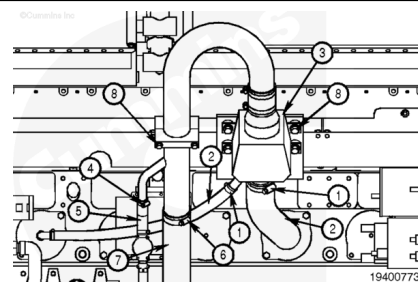


Install the breather box assembly.

- Connect the breather box to the harness support bracket with six bolts (8).

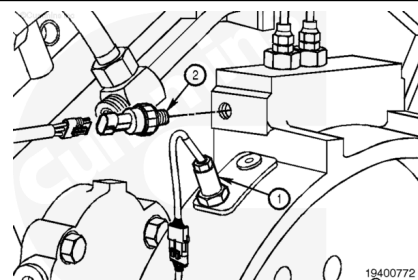
Torque Value: 45 n·m [33 ft-lb]

- Connect the lower breather hose (7) with the hose clamp (6).
- Connect the blowby pressure sensor hose (5) with the hose clamp (4).
- Connect the three hoses (2) on the bottom of the breather box (3) with the three hose clamps (1).



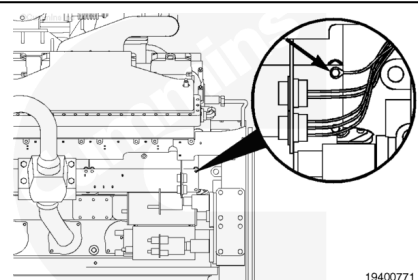
Connect the engine speed sensor (1) to the engine harness connector.

Connect the oil pressure sensor (2) to the engine harness connector.

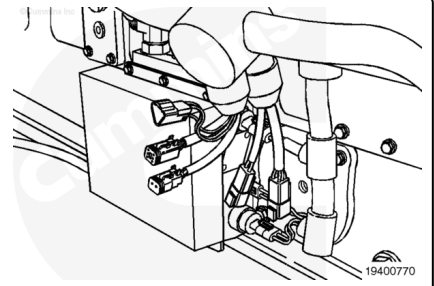


Connect the engine block ground.

Torque Value: 20 n·m [177 in-lb]

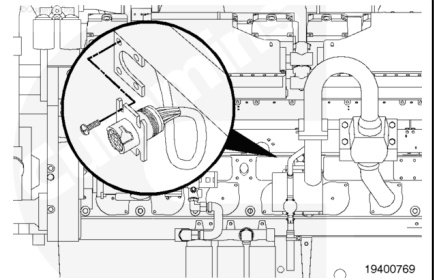


Connect the three data link connectors.



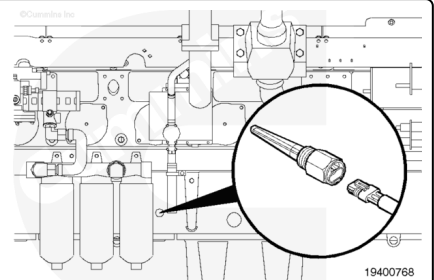
Connect the 9-pin data link connector to the engine harness support bracket.

- Slide the 9-pin connector into the engine harness support bracket connector slot.
- Secure the 9-pin connector with four screws.



Torque Value: 1 n•m [9 in-lb]

Connect the oil level sensor to the engine harness.

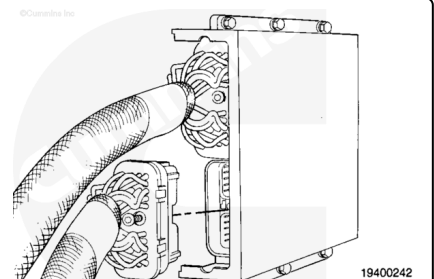


Connect the engine harness ECM connector to the ECM.

Use an inch-pound torque wrench, Part Number 3376592, with 4 mm [5/32 in] hex head adapter to tighten the connector jackscrew.

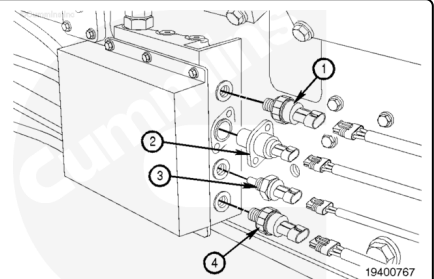
Torque Value: 2.8 n•m [25 in-lb]

Note : Do not over-torque as connector damage can occur.



Connect the engine harness to the following:

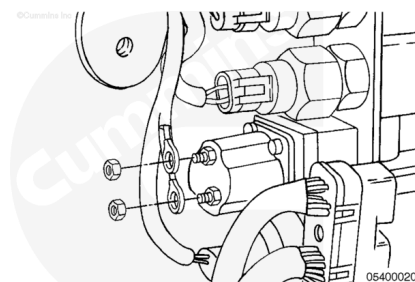
- Timing pressure sensor (1).
- Ambient air pressure sensor (2).
- Fuel temperature sensor (3).
- Rail pressure sensor (4).



Connect the fuel shutoff control wires and ground wires to the fuel shutoff solenoid.

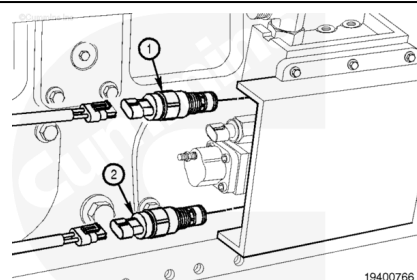
Tighten the retaining nuts.

Torque Value: 3 n•m [27 in-lb]



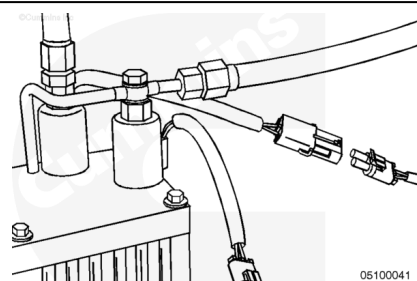
Connect the timing actuators (1) to the engine harness.

Connect the rail actuator (2) to the engine harness.



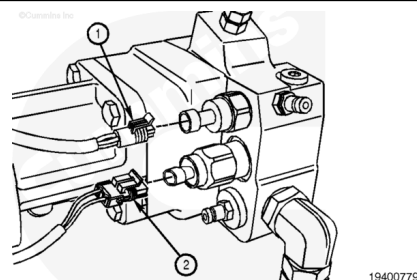
If equipped, connect the Centinel™ burn valve solenoid to the engine harness.

If equipped, connect the Centinel™ make-up valve solenoid to the engine harness.

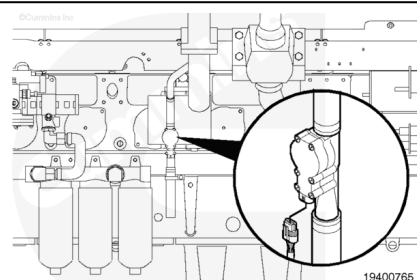


Connect the fuel pump pressure sensor (1) to the engine harness.

Connect the fuel pump actuator (2) to the engine harness.

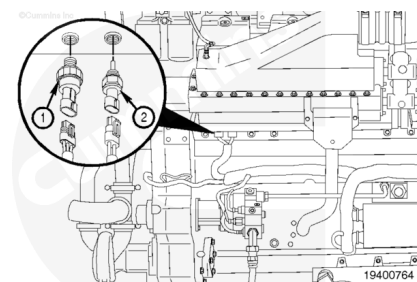


If CENSE™ equipped, connect the blowby pressure sensor to the engine harness.



Connect the intake manifold air pressure sensor (1) to the engine harness.

Connect the intake manifold air temperature sensor (2) to the engine harness.



Connect the coolant temperature sensor (1) to the engine harness.

If CENSE™ equipped, connect the coolant pressure sensor (1) to the engine harness.

